

Phase 5: Project Development Phase

Number of Functional Features Included in the Solution

1. Functional Features Catalog

ComicCraft delivers a comprehensive suite of twelve (12) integrated functional features spanning user input capture, AI multimodal orchestration, sequential layout building, visual preview, and document publishing:

Feature #	Feature Name	Associated Module / File	Description & Capability
F-01	Multi-Attribute Story Input Form	templates/index.html, app/routes.py	Captures Story Prompt, Character Name, Setting, Story Tone, and Art Style with validation.
F-02	1-Click Story Presets	static/js/main.js, templates/index.html	Instant auto-fill for 4 curated story concepts (The Brave Fox, Cyberpunk Detective, Cosmic Odyssey, Magic Academy).
F-03	Gemini 1.5 Flash 5-Panel Outliner	app/gemini_flash.py	Automatically structures any narrative prompt into a 5-panel JSON arc with panel numbers, titles, scene descriptions, and image prompts.
F-04	Gemini 1.5 Pro Narrative Expander	app/gemini_pro.py	Enriches outlines into comic-style stories with environmental captions, narrator voice, and character dialogues.
F-05	Diffusion Visual Inking Engine	app/image_generator.py	Synthesizes stylized 768x512 comic illustrations aligned with user-selected art styles (Anime, Comic Book, Realistic, Pixel Art, Cyberpunk).
F-06	Concurrent Multi-Threaded Rendering	app/routes.py (ThreadPoolExecutor)	Generates all 5 panel illustrations simultaneously, reducing processing time by 80%.
F-07	Panel Caching & Sanitization	app/image_generator.py	Applies MD5 prompt hashing and slugification to prevent duplicate renders and ensure deterministic file operations.
F-08	Sequential Layout Binding	app/layout_builder.py	Matches images, story text, captions, and descriptions into structured panel dictionaries.
F-09	Interactive Comic Preview Interface	templates/comic_preview.html	Renders a sequential comic strip with custom speech bubbles, yellow caption banners, dark narrator cards, and prompt references.
F-10	Publication-Ready PDF Compiler	app/exporters.py	Assembles a clean 5-page PDF document using FPDF2 with TrueType Unicode fonts (DejaVuSans.ttf), centering art and story cleanly on 1 page per panel.
F-11	Export Confirmation & Sharing Flow	templates/export_success.html, app/routes.py	Auto-redirects to a celebratory export success page with re-download capability and a call-to-action to create another comic.
F-12	Headless REST API (/generate-comic/json)	app/routes.py	Exposes a Pydantic-validated JSON API endpoint enabling third-party and programmatic comic creation.

2. Feature Coverage Verification

All functional features have been implemented, integrated, and validated across unit tests and end-to-end browser workflows.